

Deploying Web App on ECS – Step 2: Configure Security Group

To Begin with the Lab

Summary of the Lab

In this step, two **security groups** were created to control traffic in the ECS architecture. One security group was created for the **Application Load Balancer** so it could accept HTTP traffic from the internet on port 80, while the other security group was created for the ECS task so it could accept only traffic coming from the load balancer. Outbound traffic was left open so the task could still reach services like **Amazon ECR**, **Amazon S3**, and the internet through the NAT setup. This design kept the ECS task private while still allowing the application to be reached safely through the load balancer.

Understanding Security Groups in ECS Networking

A **security group** is a virtual firewall that controls what traffic can reach a resource in **AWS**. It exists so you can allow only the traffic that is actually needed instead of exposing your application to the whole internet. In this lab, one security group protects the **Application Load Balancer**, and the other protects the ECS task by allowing traffic only from the load balancer security group. The benefit is a safer architecture where users can access the application through the load balancer, but the ECS task itself stays protected inside the private subnet.

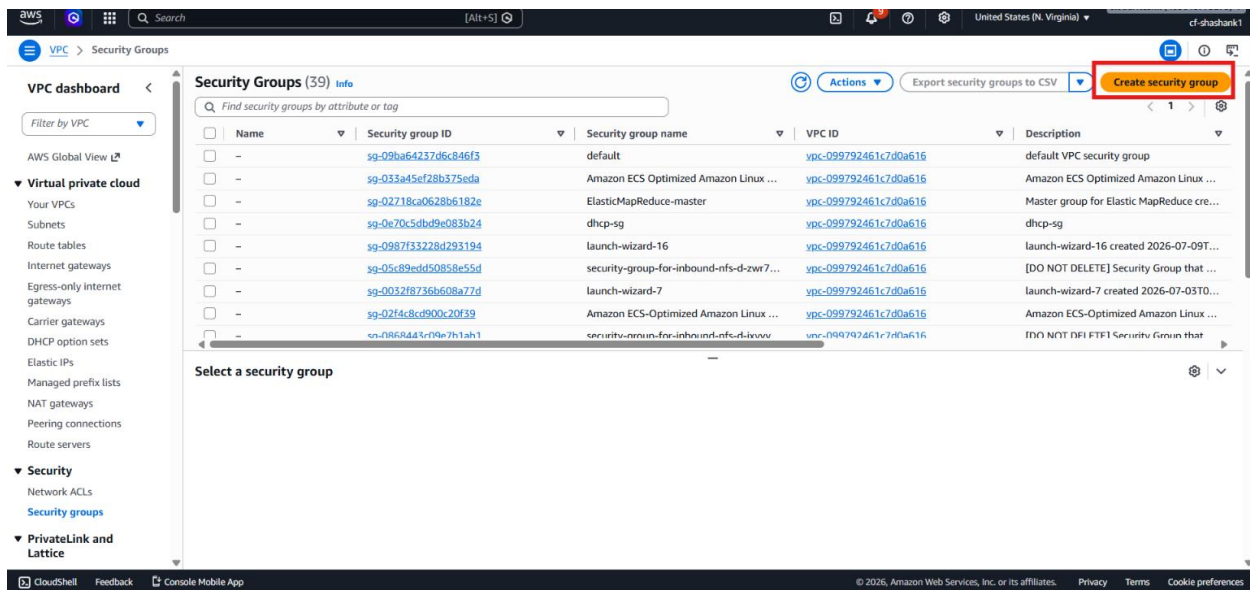
Example:

A user opens the website from a browser, so the request reaches the **Application Load Balancer** on port 80. The load balancer then forwards the request to the ECS task, but only because the task security group allows traffic from the load balancer security group. If any other system tries to reach the task directly, the request is blocked.

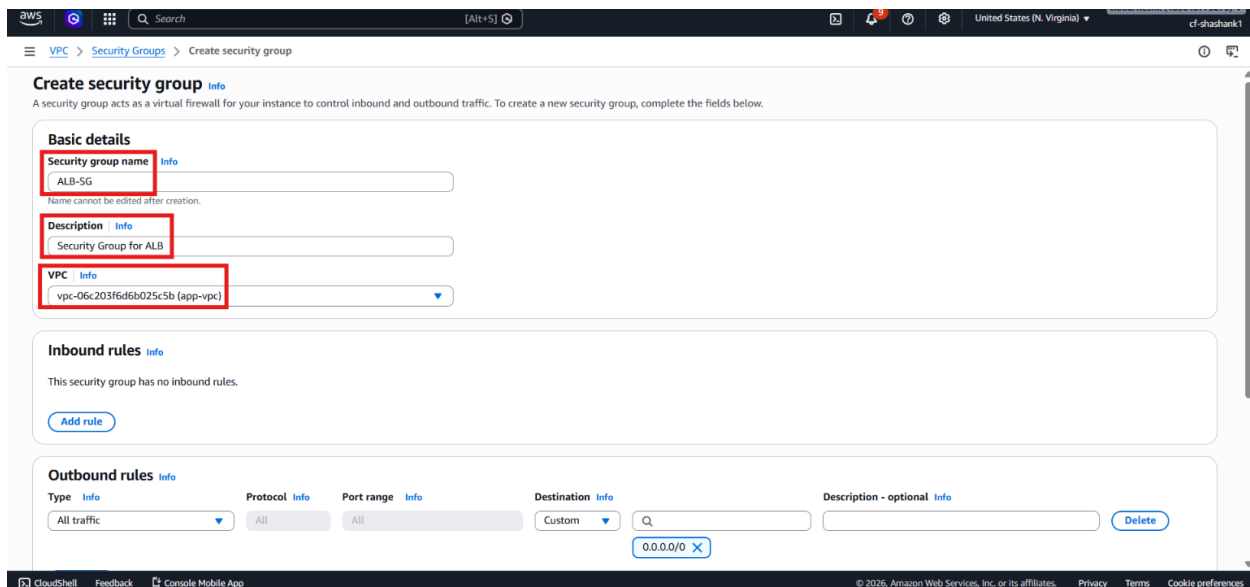
Lab Steps

Step 1 – Create the security group for the Application Load Balancer

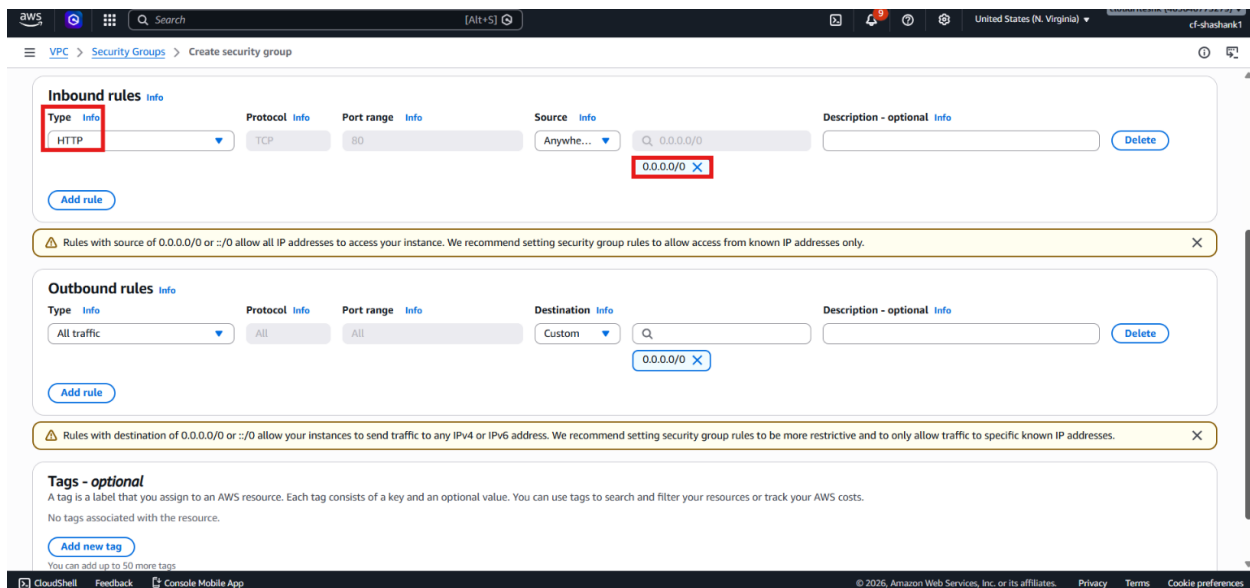
- Sign in to the **AWS Management Console**.
- Open **Amazon VPC** and go to security groups.
- Click **Create security group**.



- Enter the name: ALB-SG.
- Add the description: Security Group for ALB.
- Select the VPC we have created earlier.

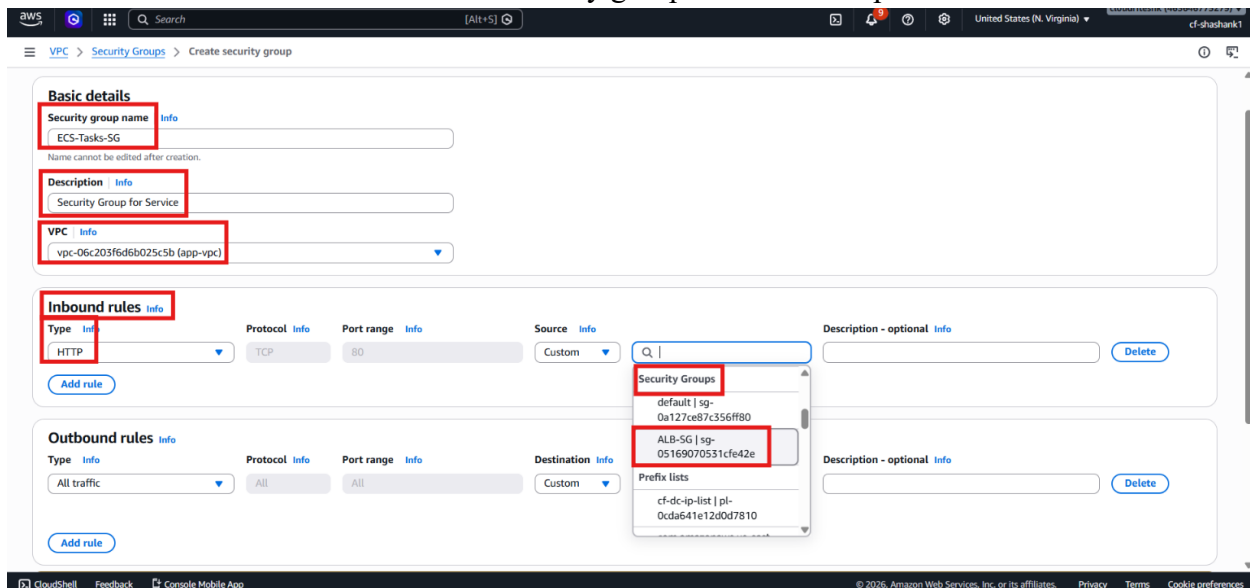


- Add an inbound rule for **HTTP**.
- Allow HTTP from 0.0.0.0/0.
- Leave outbound rules open.
- Click **Create security group**.



Step 2 – Create the security group for the ECS task

- Click **Create security group** again.
- Enter the name: ECS-Tasks-SG.
- Add the description: Security Group for Service.
- Select the same VPC.
- Leave outbound rules unchanged.
- Add an inbound rule for **HTTP** on port 80.
- Set the source to the load balancer security group instead of the public internet.



- Click **Create security group**.

Step 3 – Keep the default security group untouched

- Do not use the default security group for this setup.
- Keep the new security groups separate for cleaner control and better practice.
- Use the load balancer security group for internet traffic and the ECS task security group for private traffic only.

Verification

- Verified that the security group for the **Application Load Balancer** was created successfully.
- Verified that the load balancer security group allows **HTTP** traffic from the internet.
- Verified that the ECS task security group was created successfully.
- Verified that the ECS task security group allows **HTTP** traffic only from the load balancer security group.
- Verified that outbound access remained open for the ECS task.
- Verified that the default security group was not used for this architecture.
- Verified that the network design is ready for ECS service and load balancer setup.